

Primary Effusion Lymphoma in an Immunocompetent Elderly Patient

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Background

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Primary effusion lymphoma (PEL) is a rare and aggressive subtype of B-cell non-Hodgkin lymphoma characterized by malignant effusions without a detectable solid tumor mass. It is classically associated with immunocompromised states, particularly Human Herpesvirus 8 (HHV-8) and HIV infection. However, rare cases in immunocompetent individuals have been increasingly recognized and are often misattributed to more common etiologies such as congestive heart failure or infection. Given its poor prognosis, with a median survival of less than 6 months, early recognition is critical.

Case Description

An 89-year-old female with a history of ischemic heart failure with reduced ejection fraction (HFrEF, EF 30-35%) and atrial fibrillation (AF) presented with progressive dyspnea and orthopnea, representing her fourth hospitalization in six months for recurrent pleural effusions. Prior admissions were attributed to acute decompensated heart failure (ADHF), supported by elevated BNP and consistent symptomatic improvement following IV diuresis and thoracentesis. On this admission, chest X-ray demonstrated a large left-sided pleural effusion, similar to prior imaging. CTA chest confirmed a large left and moderate right pleural effusion with compressive atelectasis, without pulmonary embolism or mass. Initial labs revealed WBC 13.8, Hgb 12.9, Na 131, Cr 0.9, BNP 512 (previously 1200s), and hs-troponin 21 with minimal delta. During prior hospitalizations, pleural effusions were drained with transient improvement; however, pleural fluid cytology was not obtained, and the effusions were presumed to be secondary to HFrEF. Given the recurrent, unilateral predominance and discordance with expected ADHF physiology, a more comprehensive pleural fluid evaluation was pursued.

Case Description Contd.

Cytology subsequently revealed malignant lymphoid cells, and flow cytometry demonstrated a monoclonal B-cell population (CD19+, CD20+, CD10 dim+, lambda restricted).

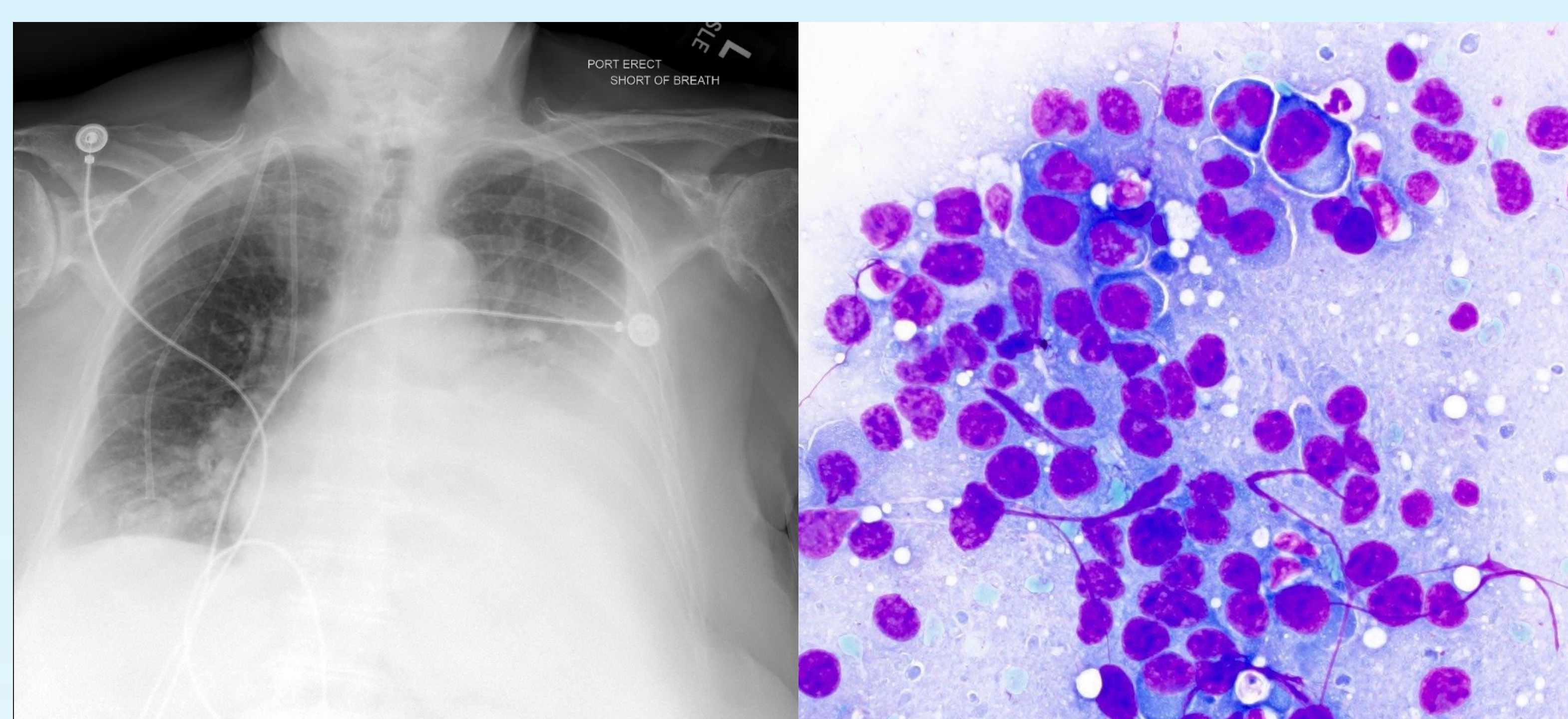


Figure 1. Chest radiograph of the patient demonstrating a large left-sided pleural effusion with associated compressive atelectasis (left) Pleural fluid cytology demonstrating large atypical lymphoid cells with prominent nucleoli and basophilic cytoplasm (right)

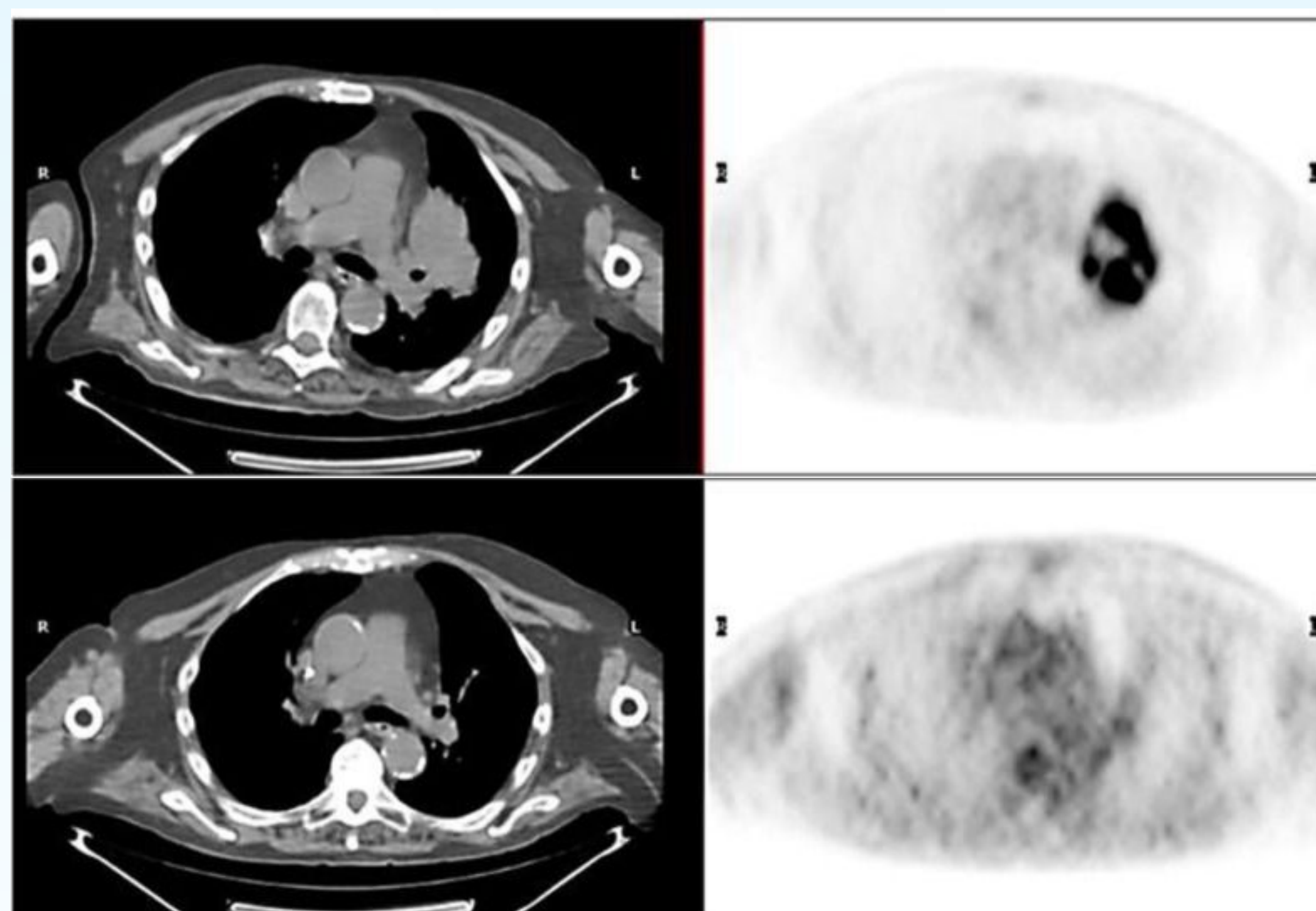


Figure 2. FDG PET-CT prior to treatment (top) and response following R-mini-CHOP therapy (bottom)

Clinical Pearls

Not all effusions are benign: Always consider cytologic evaluation.

PEL can mimic CHF: Recurrent effusions despite aggressive diuresis should raise suspicion

Cytology saves lives: Routine pleural cytology is often skipped

PEL is not exclusive to the immunocompromised: Particularly the elderly.

Early diagnosis enables intervention: In an otherwise rapidly fatal disease.



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